Victor Boussange
Applied ML Researcher, Ph.D.

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Lausanne, Switzerland   French citizen
227 GitHub stars earned   215 citations

Updated as of July 7, 2026.

I am an applied ML researcher working in AI for science. I develop robust methods for high-dimensional, spatio-temporal forecasting and inverse problems that remain reliable under low signal-to-noise ratios and distribution shift, with applications in biology and environmental sciences. I am particularly interested in hybrid modelling, integrating domain priors with deep learning and GPU-accelerated differentiable computing. I ship open-source libraries implementing these methods, lead funded research projects, and mentor students in applied machine learning and scientific computing.

Technical skills & open-source contributions

Programming languages	**Python**, **Julia**, C++, Java, MATLAB, R, Bash, SQL, LaTeX.
ML & deep learning	**PyTorch**, **JAX**, scikit-learn, Lux.jl; neural differential equations, geospatial deep learning, graph neural networks, differentiable programming, dynamical systems & time-series forecasting, hybrid/mechanistic modelling, Bayesian inference.
Scientific & data engineering	(sparse) linear algebra, numerical simulations of partial differential equations, geospatial and molecular data processing pipelines, distributed computing, Slurm HPC, Xarray.
Open-source contributions	Member of the SciML organisation (scientific machine learning), with contributions to SciML core packages, JAX, lineax, CUDA.jl, Flux.jl, and Graphs.jl.

Professional appointments

Present – December 2025
Research fellow, EPFL, School of Computer and Communication Sciences, Switzerland. Developed differentiable sparse linear solvers for Laplacian systems with application to graph learning. Currently developing GNNs for forecasting nonlocal spatio-temporal dynamics with application to disease and pest spread. Supervised 4 B.S./M.S. students in computer science.
graph neural networks | spatio-temporal forecasting | team supervision

November 2025 – November 2022
Postdoctoral researcher, ETH Zürich | Swiss Federal Institute for Forest, Snow and Landscape (WSL), Switzerland. Built multi-scale geospatial deep learning models for biodiversity estimation, improving accuracy by 61% over state-of-the-art methods. Developed a GPU-accelerated, differentiable open-source library for ecological connectivity assessment, deployed at national scale in Switzerland. Co-supervised 2 Ph.D. students.
geospatial deep learning | graph neural networks | scientific software engineering

August 2018 – March 2018
R&D intern, Compagnie Nationale du Rhône (CNR), France. Built optimisation software for energy trading and virtual power plant management.
software development | mathematical optimisation | energy markets

Education

October 2022 – September 2018
Ph.D. in Applied Mathematics and Environmental Sciences, ETH Zürich, Switzerland. Developed numerical methods for efficiently solving high-dimensional nonlinear PDEs arising in biology, finance, and engineering, and for calibrating them against time-series data. Advised by Prof. Dr. Loïc Pellissier, Prof. Dr. Didier Sornette, and Prof. Dr. Arnulf Jentzen.
scientific machine learning | high-dimensional PDEs | biomathematics

June 2017 – February 2017
Master's thesis, CSIRO, Australia. Numerical continuation and bifurcation analysis applied to nonlinear dynamical systems. Advised by Dr. Thomas Poulet.
scientific computing | dynamical systems | numerical methods

August 2018 – September 2013

B.S./M.S. in Energy and Environmental Engineering, INSA Lyon, France.

- Two-year intensive undergraduate course in mathematics and physics. Ranked 21st/650 students.
- Three-year engineering curriculum in Energy and Environmental Systems.

optimisation | energy markets | fluid mechanics | thermodynamics

Selected software

AMJAX 2026
github.com/vboussange/AMJAX
Algebraic multigrid solvers in JAX.
ML library

MuScaRi 2026
github.com/vboussange/MuScaRi
Implementation and pre-trained weights for MuScaRi, a model for multi-scale richness estimation.
Model

HybridDynamicModels.jl 2025
github.com/vboussange/HybridDynamicModels.jl
`Lux.jl` layers and utilities to build and train hybrid state-space models.
ML library

JAXScape 2024
github.com/vboussange/jaxscape
A GPU-accelerated and differentiable library for landscape genetics and ecological connectivity modelling.
Domain-centric library

HighDimPDE.jl 2021
github.com/SciML/HighDimPDE.jl documentation
A Julia package that breaks down the curse of dimensionality when solving nonlocal, nonlinear PDEs.
ML library

Selected publications

Preprints

- Plekhanova, E., Brun, P., Fischer, J.-C., Ma, H., **Boussange, V.**, Zimmermann, N. E., *Clustering biomes from space: from pixels to foundation models*. [SSRN] (2026).
- **Boussange, V.**, Brun, P., Malle, J. T., Midolo, G., Portier, J., Sanchez, T., Zimmermann, N. E., Axmanova, I., Bruelheide, H., Chytry, M., Kambach, S., Lososova, Z., Vecera, M., Birrun, I., Ecker, K. T., Lenoir, J., Svenning, J.C., Karger, D. N., *Multi-scale species richness estimation with deep learning*. [arXiv] (2025). Revision requested by Nature Communications.

Peer-reviewed

- Dieing, M., Volpi, M., **Boussange, V.**, *Biodiversity Profile Estimation with Earth Observation*. Accepted at the IAPR Workshop on Pattern Recognition in Remote Sensing (PRRS 2026).
- **Boussange, V.**, Vilimelis-Aceituno, P., Pellissier, L., *A calibration framework to improve mechanistic forecasts with hybrid dynamic models*. Methods in Ecology and Evolution (2025).
- Sapienza, F., Bolibar, J., Schäfer, F., Groenke, B., Pal, A., **Boussange, V.**, Heimbach, P., Hooker, G., Pérez, F., Persson, P.O., Rackauckas, C., *Differentiable Programming for Differential Equations: A Review*. [arXiv] (2024), 72 pages. Accepted at SIAM Review.
- Aceituno, P., Miller, J., Marti, N., Farag, Y., **Boussange, V.**, *Temporal horizons in forecasting: a performance-learnability trade-off*. Transactions on Machine Learning Research (2025).
- **Boussange, V.**, Becker, S., Jentzen, A., Kuckuck, B., Pellissier, L., *Deep learning approximations for non-local nonlinear PDEs with Neumann boundary conditions*. Partial Differential Equations and Applications, Paper no. 51, 59 pages (2023). [arXiv]
- Skeels, A., Boschman, L. M., McFadden, I. R., Joyce, E.M., Hagen, O., Jiménez Robles, O., Bach, W., **Boussange, V.**, Keggin, T., Jetz, W., Pellissier, L., *Paleoenvironments shaped the exchange of terrestrial vertebrates across Wallace's Line*. Science 381, 86-92 (2023).

Supervision and Teaching

2026	**Semester project supervision, EPFL, IC, Switzerland.** Ghali Elouahdani, ”Accelerating ecological connectivity analysis with graph neural networks”.
2026	**Semester project supervision, EPFL, IC, Switzerland.** Fanny Missier, ”Developing an algebraic multigrid solver in JAX”.
2025	**Master’s student supervision, ETH Zürich, D-INF, Switzerland.** Jeffrey Zweidler, ”Forecasting invasive species range expansion using ecologically informed neural networks”, in collaboration with the Swiss Data Science Center.
2025	**Master’s student supervision, ETH Zürich, D-INF, Switzerland.** Moritz Dieing, ”Attention-based deep multiple instance learning for species richness prediction”, in collaboration with the Swiss Data Science Center.
2024-2028	**Ph.D. student co-supervision, ETH Zürich, D-USYS, Switzerland.** Capucine Lechartre, ”A mechanistic approach to biome transitions across space and time”.
2023-2024	**Lecturer, Environmental Sciences Master, ETH Zürich, D-USYS, Switzerland.** Course 701-1679-00L, Landscape Modelling of Biodiversity: From Global Changes to Conservation.
2023	**Lecturer, Environmental Sciences Bachelor, ETH Zürich, D-USYS, Switzerland.** Course 701-3001-00L, Environmental Systems Data Science. Responsible for the unit *Supervised Deep Learning: Application*.
2020	**Lab rotation supervision, ETH Zürich, D-BSSE, Switzerland.** Cecilia Valenzuela Agui, MS *Computational Biology and Bioinformatics*.
2020	**Intern supervision, Polytech Nice-Sophia, France.** Nicolas Demolin, Taste of Research internship, MS *Applied Mathematics and Modeling*.

Funding

2026 (pending)	**500,000 CHF**	International Joint Initiative for Research Harnessing Disruptive Technologies to Address Global Challenges; co-PI on ***GreenTwin****: Participatory Digital Twins for biodiversity dynamics and sustainable land-use planning*. Coordinator: Bram Van Moorter.
2025	**100,000 CHF**	SNSF Spark grant for ***EcoDynCast****: forecasting alien invasive species range dynamics with GNNs*.
2024	**50,000 CHF**	*A course on reproducible research, data pipelines, and scientific computing*. Coordinator: Mauro Werder.
2024	**10,000 EUR**	yDiv Graduate School & Postdoc Program workshop funding for a three-day event.
2023	**10,000 CHF**	WSL Biodiversity Center Innovative Workshop Grant.

Languages

French	●●●●●	Spanish	●●●○○
English	●●●●●	German	●●●○○

Hobbies

- **Mountain sports.** Swiss Alpine Club tour leader; ski mountaineering, alpinism, rock climbing, and enduro mountain biking.
- **Sailing and expeditions.** Refit a 36-foot, 40-year-old sailing boat and sailed it from Germany to Norway; long-distance ski/bicycle traverses in Norway and the Alps.
- **Video editing.** ”Vagabonds du Grand Nord”, official selection at Xplore Alps Festival 2025

See full adventure CV for details.

References

Dr. Michele Volpi	**Prof. Dr. Niklaus Zimmermann**	**Prof. Dr. Loïc Pellissier**
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